CS 499 Module One Assignment

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I. Enhancement Plan:

A. **Category One:** Software Engineering and Design
   i. **Select an artifact** that is **aligned with the** software engineering and design **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan.

General Artifact Overview

Artifact: Events Android App

Origin: CS-360 Mobile Architecture and Programming

The initial project consists in a user-centered Android application developed using Android Studio IDE, Java programming language, and SQLite. This project emphasizes modern UI/UX best practices, lifecycle-aware architecture, secure authentication, and seamless performance, all while reducing user cognitive load and delivering a smooth, distraction-free experience.

This application was designed to function as a distraction-free personal scheduling hub that empowers users to manage multi-day commitments with clarity and control. It presents events in an agenda-style list view, grouped by date and sorted in ascending chronological order, allowing users to quickly understand their upcoming schedule and stay organized with minimal effort.

Main Features:

- Secure Login and Sign-Up authentication system.
- Secure password storage using Bcypt hasing and salting.
- Native SQLite database for users and events.
- Create and edit events functionality through a form.
- Smart and frictionless recurring event generation logic (start date + optional end date), with flexibility for the users to edit individual iterations.
- Chronologically grouped event dashboard.
- Daily SMS notification summaries.
- Android WorkManager for reliable SMS task scheduling.
- Background threads for database and SMS operations.
- Model–View–ViewModel (MVVM) architecture for separation of concerns and preserve UI state across configuration changes.
- LiveData and Observers for real time local updates.

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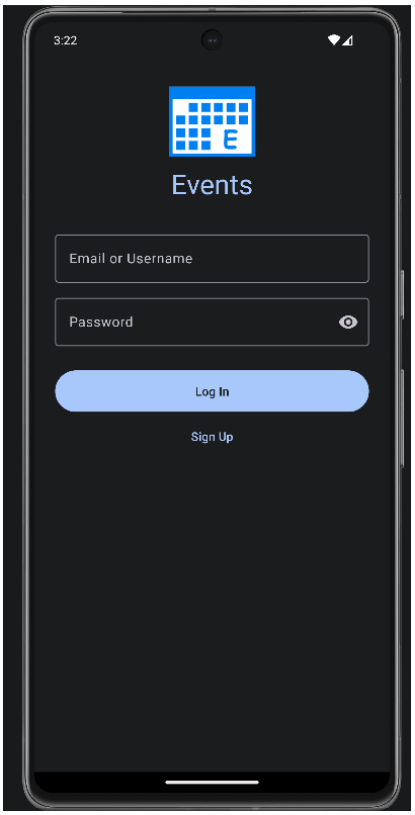
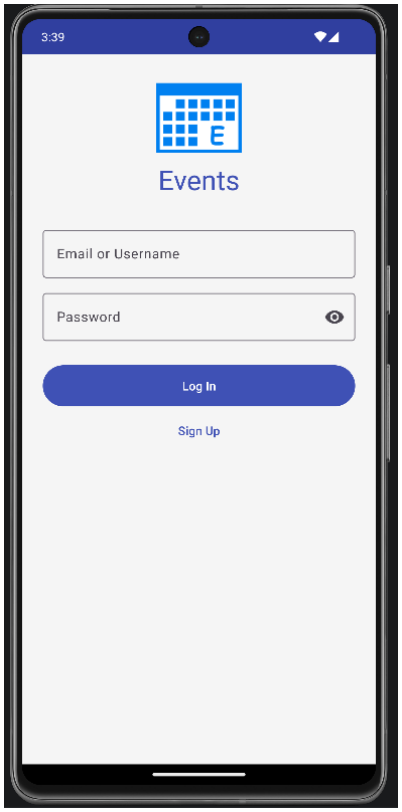
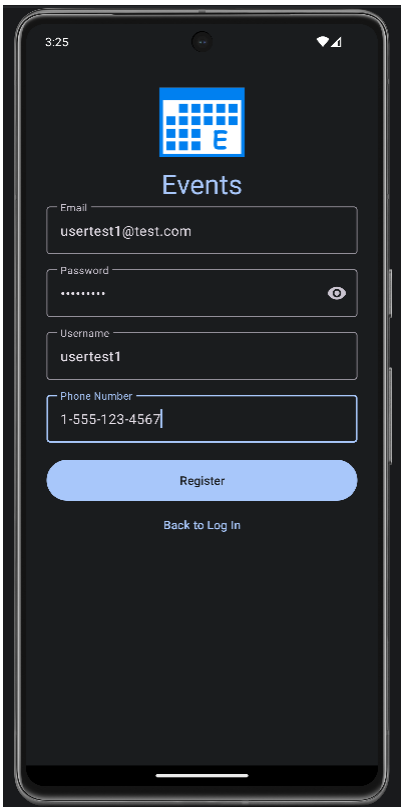
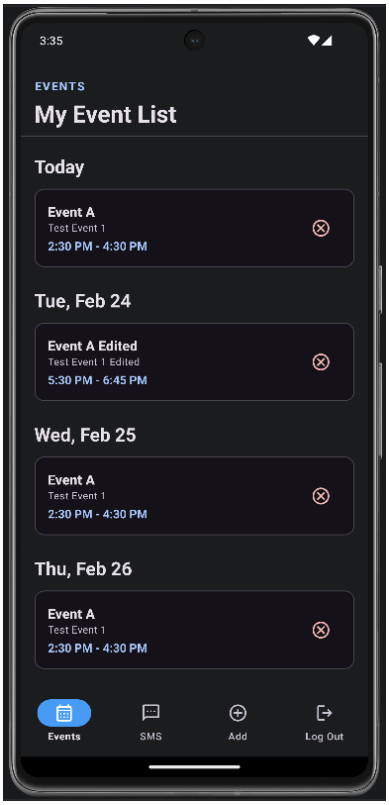
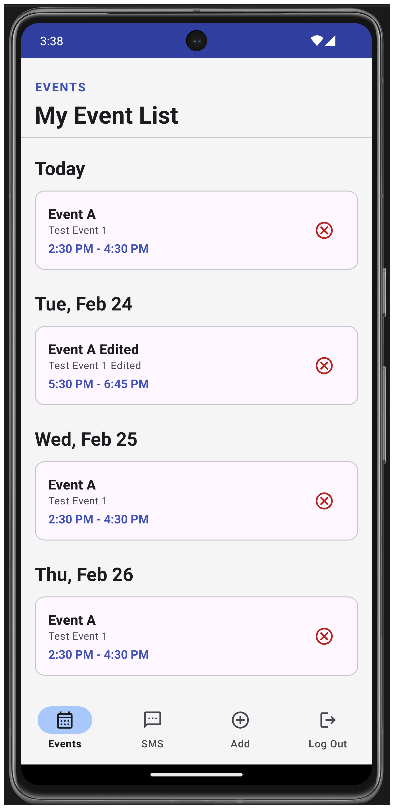
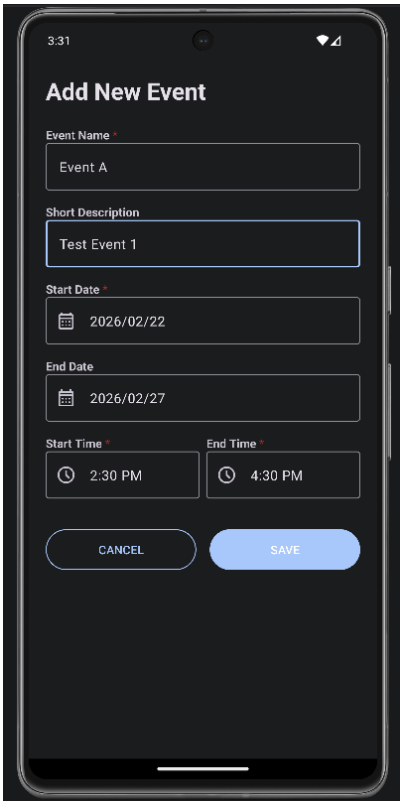












ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

Software Engineering and Design Enhancements

In the initial submission, I implemented basic login security through user authentication and Bcrypt hashing for password management. However, several industry-standard security practices were still pending implementation, such as multifactor authentication, limiting login attempts, password recovery options, and stateless authentication to maintain user sessions securely. From a functionality perspective, the application currently displays events in a chronological list grouped by date, which provides a clear agenda-style overview. To further improve the user experience, I plan to add a calendar view that visually displays events across multiple days. Additionally, past events currently remain visible in the main list; to reduce cognitive load and keep the interface focused on relevant information, these events will be grouped into a collapsible dropdown section at the bottom so that current-day events remain prioritized. To summarize, I plan to include the following enhancements in this category:

- Multifactor authentication using SMS verification with the user's registered phone number.
- Limiting login attempts to prevent brute-force attacks.
- Password change and recovery using MFA SMS verification codes.
- Stateless authentication using token-based sessions that remain active until explicit logout.
- A calendar view to complement the chronological agenda list.
- Grouping past events into a dropdown section to keep the main view focused on current and upcoming events.

Implementation Plan

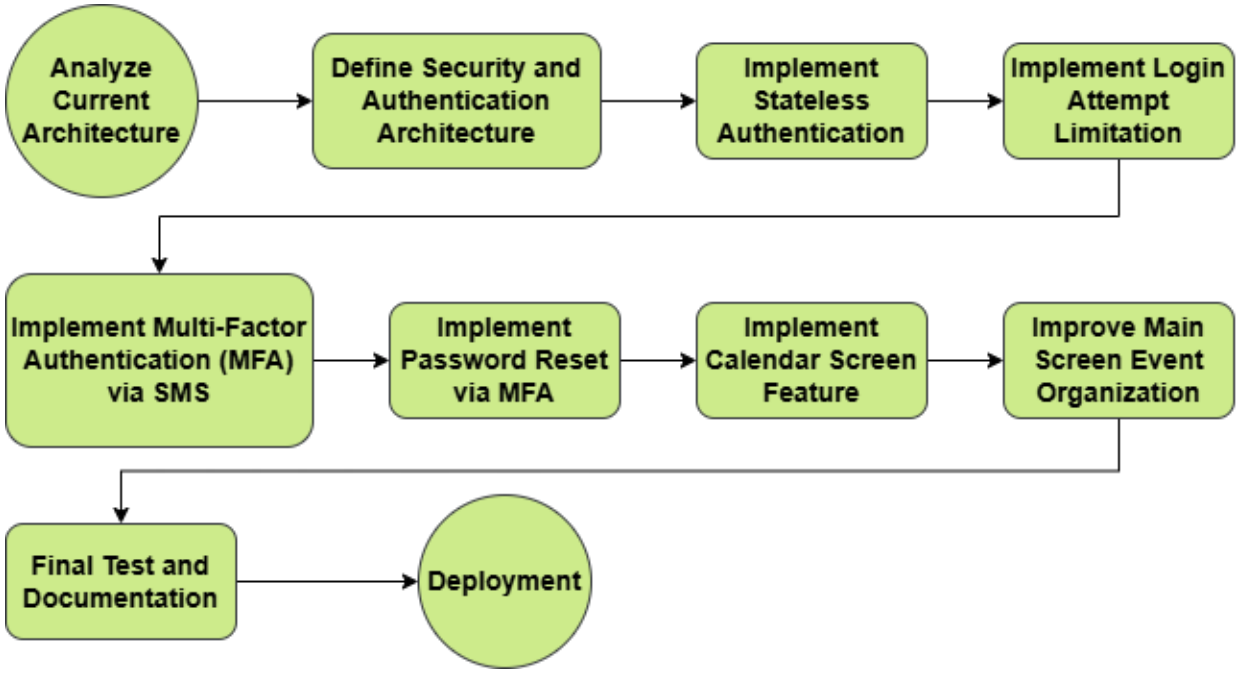
Firstly, I will analyze the current architecture to identify any structural limitations and areas that may require refactoring before implementing additional security features. In this phase, I will review the existing authentication and session management processes to understand how user credentials and login attempts are currently handled, as well as evaluate the UI architecture of the login screen. Based on this assessment, I will design an improved security and authentication architecture that supports secure and scalable session management. Specifically, I will define a token-based stateless authentication model using session tokens such as JSON Web Token (JWT), and establish a complete authentication flow that includes login validation, multifactor verification, and password recovery mechanisms. After defining this architecture, I will modify the login process to generate secure authentication tokens upon successful login, store them securely, validate them for each authenticated request, and ensure sessions persist until the user explicitly logs out. I will also implement login attempt validation to mitigate brute-force attacks by tracking failed login attempts per user, defining a threshold, and redirecting users to the account recovery process when necessary.

Next, I will implement the core feature that enables secure recovery and verification: multifactor authentication. This will be implemented via SMS by extending the login workflow to require an additional verification step after password validation. The system will generate a temporary one-time verification code and send it to the user's registered phone number via SMS. Users will then need to input this code to complete authentication and gain access. After implementing MFA, I will develop a

password reset feature that uses the same SMS verification process, allowing users who exceed the login attempt threshold to recover their account only after confirming their identity with a unique verification code. In addition to security enhancements, I will improve the user experience by implementing a calendar interface on a separate screen to complement the existing agenda view. This will involve adding a calendar icon on the bottom navigation bar, connecting the calendar view to the database events table, enabling navigation across different time periods, and displaying event summaries when a user selects a specific date. Finally, I will refactor the main events list to prioritize upcoming events and reduce cognitive load by filtering visible elements and grouping past events under a collapsible dropdown section at the bottom.

Throughout the development process, I will follow structured version control practices using Git and GitHub. I will create dedicated branches for each feature, documenting bug fixes, new implementations, and refactoring changes through descriptive commit messages. In addition, I will document functions, file scopes, and overall implementation decisions within the codebase and supporting project documentation, including capstone journals and technical artifacts. Each feature will be thoroughly tested before merging into the main branch, ensuring that every stage of development is validated and stable before progressing to the next release milestone.

Implementation Flowchart



iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate that align with the course outcome.

In this enhancement, I will demonstrate the following skills:

- Secure authentication system design.
- Stateless architecture implementation.
- Java programming language.
- System security design and threat mitigation.
- Modular software architecture.
- UI/UX design.
- Multi-factor authentication.
- Software security.
- Protection against brute-force attacks.
- Token-based authentication.
- Architectural system design.
- Android components and layout implementation.

b. Select one or more of the course outcomes below that your enhancement will align with.

This enhancement aligns with the first outcome because it incorporates strategies for building in collaborative development environments through the use of version control, structured documentation, and repository management using GitHub. It also supports the second outcome by requiring the design, development, and delivery of professional-quality technical communication, including clear code comments, a well-structured repository README, and detailed capstone documentation artifacts that explain implementation decisions and system behavior.

Additionally, the enhancement supports the third outcome by applying core Computer Science principles to design and evaluate secure computing solutions. This includes designing secure authentication workflows and assessing architectural trade-offs involved in implementing token-based session management and multifactor authentication. For the fourth outcome, the project applies well-founded and modern development techniques and tools, including calendar interface components, stateless authentication mechanisms using tokens such as JSON Web Token, and other contemporary implementation strategies.

Finally, the enhancement aligns with the fifth outcome by demonstrating a strong security-focused mindset throughout the design process. By anticipating potential vulnerabilities, the system will implement safeguards such as limiting login attempts to prevent brute-force attacks, integrating multifactor authentication, and managing authentication tokens securely.

B. **Category Two:** Algorithms and Data Structures

i. **Select an artifact** that is **aligned with the** algorithms and data structures **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan. You may choose work from the courses listed under Category One.

Artifact: Events Android App

Origin: CS-360 Mobile Architecture and Programming

For this category, I will use the same mobile application artifact and enhance it by implementing search and sorting functionalities while applying appropriate data structures and algorithms to optimize time and space efficiency. This enhancement will ensure that event retrieval and organization are performed quickly and efficiently, even as the dataset grows.

ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

Algorithms and Data Structures Enhancements

The initial application displays events in a list grouped by date in ascending chronological order. However, in its current form, users must manually scroll through the entire list to locate a specific event. To improve usability and efficiency, I will enhance the application by implementing search functionality that leverages appropriate data structures and algorithms to improve data retrieval performance.

This feature will be implemented in a separate screen where users can search for events by matching keywords in the event title or description. Once a query is submitted, the application will return a list of matching results sorted in ascending chronological order. Each result will display key event information, including the title, description, date, start time, and end time, allowing users to quickly identify the relevant event without manually navigating through the full agenda.

To summarize, I plan to include the following enhancements in this category:

- Events search algorithm by matching keywords.
- Sorting algorithm to complement the search feature.
- Events data structure refinement to implement search and sorting algorithms.

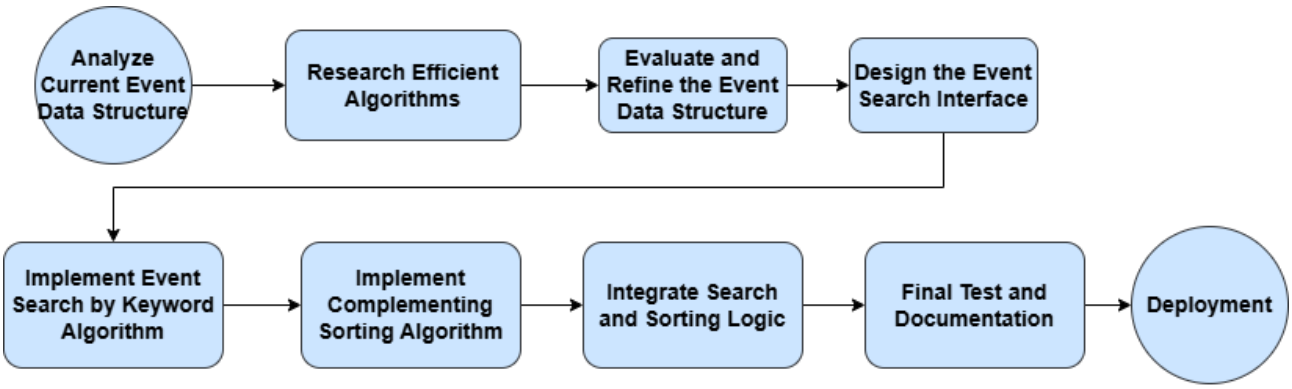
Implementation Plan

Firstly, I will analyze the current event data structure to identify any limitations affecting sorting and retrieval performance. Based on this analysis, I will research efficient search and sorting algorithms that are appropriate for mobile environments, where computing resources are limited and both time and space efficiency are important. During this phase, I will compare different keyword matching techniques for searching event titles and descriptions, evaluate their Big O complexity, and select the approach that best balances performance and simplicity for the application's dataset size and usage patterns.

Once the algorithms are selected, I will evaluate and refine the event data model to ensure it supports efficient searching and sorting operations. After finalizing these data structure decisions, I will design the event search interface by adding a dedicated screen for event lookup and including the corresponding icon in the bottom navigation bar. This screen will allow users to input search queries and view results in a structured list format.

With the interface prepared, I will implement the keyword search algorithm followed by the complementary sorting algorithm to display matching events in ascending chronological order. Finally, I will integrate both algorithms into the application workflow and validate that results are accurately retrieved and displayed in the search interface, ensuring the feature performs efficiently and provides a smooth user experience.

Implementation Flowchart



iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate to align with the course outcome.

In this enhancement, I will demonstrate the following skills:

- Data Structures.
- Algorithms.
- Algorithmic Thinking.
- Performance Optimization.
- Software Architecture.
- UI/UX Design.
- Time and Space Complexity Analysis.
- Data Modeling.

b. Select one or more of the course outcomes listed under Category One that your enhancement will align with.

This enhancement aligns with the third outcome because it involves designing and evaluating computing solutions using core algorithmic principles. Through researching search and sorting algorithms, refining the underlying data structures, and analyzing performance trade-offs, I will apply concepts from Algorithms and Data Structures to ensure efficient event retrieval and processing within the application.

It also aligns with the fourth outcome by applying innovative techniques and technical skills to strengthen my understanding of data management and algorithmic efficiency. In particular, implementing optimized keyword lookup mechanisms will allow me to apply theoretical knowledge in a real-world context. By integrating these solutions into the application's architecture, the enhancement will demonstrate how algorithmic strategies can support scalable and responsive functionality.

Finally, this work reinforces the importance of incorporating data structures and algorithms as part of a complete system design rather than as isolated components. Integrating these elements into a

production-style mobile application architecture ensures that performance considerations, usability, and maintainability are addressed properly.

C. **Category Three: Databases**

i. **Select an artifact** that is **aligned with the** databases **category** and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan. You may choose work from the courses listed under Category One.

Artifact: Events Android App

Origin: CS-360 Mobile Architecture and Programming

For this category, I will use the same mobile application artifact and enhance it by migrating from a local relational database to a cloud-based non-relational solution. This change will improve the application's scalability and flexibility, allowing the system to handle growth in both the number of users and the volume of stored data. By moving data management to the cloud, the application will also support access across multiple devices, enabling users to retrieve and manage their events regardless of the device they are using.

ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

Database Enhancement

The initial Events application uses a local native database on the Android device through SQLite, which follows a relational database model. The current data structure includes two tables: users and events, where events are retrieved in ascending chronological order and filtered by the user_id foreign key. While this design works for a single device, it introduces a limitation: if the same user attempts to log in from a different device, the application will not recognize their credentials or event data. Even if the user creates a new account on another phone, the events list will appear empty because the data remains stored locally on each device, resulting in separate and inconsistent datasets.

To address this limitation, I will research and implement a cloud-based database solution, which is an essential skill in modern Software Engineering and Data Engineering practices. The enhancement will migrate the application from a local relational database to a distributed, document-oriented database model. This approach will replace the traditional relational schema with a flexible structure capable of supporting scalable and distributed data storage across multiple devices and users. Adopting a cloud-based database architecture will enable real-time synchronization of user data, ensuring that events and account information remain consistent regardless of the device used to access the application, improving availability, reliability, and scalability as the user base grows.

To summarize, I plan to include the following enhancements in this category:

- Migrate from local relational database to a cloud distributed, document-oriented database model.

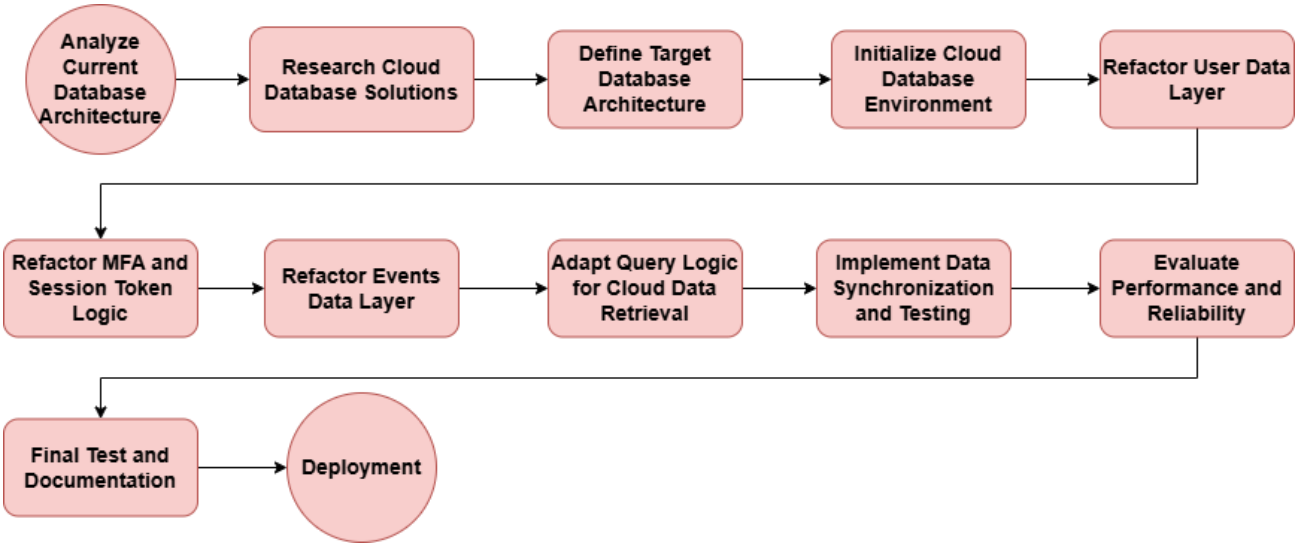
Implementation Plan

Firstly, I will analyze the current database architecture by reviewing the relational schema implemented with SQLite. This analysis will focus on understanding the existing data model and identifying its limitations in a multi-device environment. Afterward, I will research cloud database solutions capable of supporting distributed data access across multiple devices. During this phase, I will evaluate platforms such as Firebase, MongoDB, and Amazon AWS based on criteria including compatibility with Android applications, implementation complexity, cost and pricing models, server-side management requirements, scalability, and real-time synchronization capabilities. Once a solution is selected, I will define the target database architecture by replacing the rigid relational schema with a flexible data model. This will involve designing a distributed database structure based on a document-oriented NoSQL approach, determining how user and event data will be represented as documents rather than relational tables, and defining logical relationships between entities. At this stage, I will also design a data migration strategy by mapping the current relational schema to the new document-based structure and defining how existing fields will be stored within the new model.

After defining the architecture, I will initialize the cloud database environment to prepare the infrastructure required for remote data storage. This phase will include creating the database account, configuring the initial environment, and establishing secure connectivity between the mobile application and the remote database. Next, I will refactor the user data layer so that user authentication and account information are retrieved from the distributed database rather than the local device. This will require reviewing and modifying key components of the application's architecture, including the model classes, repository classes, and data access objects (DAO). Depending on the selected cloud solution, I will also determine whether it provides built-in support for security features such as multifactor authentication and password recovery. If such services are available, I will refactor the authentication logic to leverage these built-in capabilities in order to strengthen security and reliability.

Once the user data layer is refactored, I will apply a similar process to the events data layer to ensure that events remain correctly associated with their respective user accounts in the new distributed environment. I will then adapt other application features, such as the search functionality, to maintain compatibility with the previously implemented algorithms and application logic. During this phase, I will ensure that all features correctly retrieve and store data from the cloud database and that the information is organized into appropriate data structures for processing. Finally, I will implement and validate data synchronization to ensure consistent data access across devices. I will also evaluate performance and reliability by analyzing factors such as latency and response time during data retrieval and user creation operations. As with the previous enhancements, I will test each feature incrementally and document every change made throughout the development process.

Implementation Flowchart



iii. Explain how the planned enhancement will **demonstrate** specific **skills** and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate that align with the course outcome.

For this enhancement, I will migrate from a relational database model to a document-oriented database, redesign the data model to support a distributed environment, integrate the mobile application with a cloud database system, implement real-time data synchronization, design scalable data storage solutions, and manage data consistency across multiple devices. Summarizing these capabilities, this enhancement will the following skills:

- Cloud Computing
- Database Design and Architecture
- Data Modeling
- Relational Databases
- Document-Oriented Databases
- Cloud Integration
- Data Engineering
- Engineering Assessment
- Solutions Evaluation

b. Select one or more of the course outcomes listed under Category One that your enhancement will align with.

This enhancement aligns with the third outcome because it involves designing and evaluating computing solutions using core Computer Science principles to develop a new database architecture and assess the

trade-offs between relational database systems and document-oriented distributed solutions. By comparing these approaches and selecting an appropriate model for a multi-device environment, the enhancement demonstrates the application of analytical and architectural decision-making in software design. It also supports the fourth outcome by applying modern cloud-based strategies, techniques, and tools to implement scalable data management solutions. Finally, this enhancement aligns with the fifth outcome by incorporating a strong security mindset throughout the design and implementation process, ensuring secure data access across multiple devices, protecting user data during synchronization, and implementing appropriate authentication and access controls.

II. ePortfolio Overall Skill Set

A. Accurately describe the **skill set** to be illustrated by the **ePortfolio overall**.

i. Skills and outcomes planned to be illustrated in the code review

For the code reviews, I will examine my Events mobile application artifact to explain the features currently implemented across three main categories: software engineering and design, algorithms and data structures, and databases. During the review, I will identify missing components and areas that require improvement based on the assessment completed for this initial submission. I will clearly communicate these gaps and present my plan for addressing them through upcoming enhancements, demonstrating how each improvement will be implemented efficiently within the project architecture.

As part of this process, I will apply several key technical and professional skills, including code review, software project documentation, software reverse engineering and refactoring, software security, UI/UX design, authentication and multifactor authentication, data structures and algorithms, database architecture, and cloud migration. These skills will allow me to evaluate the current implementation, analyze design decisions, and propose well-reasoned solutions that strengthen the application's functionality, scalability, and security.

This deliverable will align with the second outcome by requiring the design, development, and presentation of a professional-quality code review delivered both visually and orally through a video presentation, as well as through written documentation within the codebase. It will also support the first outcome by establishing clear development guidelines that encourage collaborative work practices within a computer science environment. Finally, it will align with the third outcome by critically reviewing the existing implementation, evaluating architectural trade-offs, and proposing improvements based on best practices.

ii. Skills and outcomes planned to be illustrated in the narratives

In the narratives, I will document my research and comparison of different technologies, techniques, and tools that could be used to implement the enhancements for each category based on my selected artifacts. This process will require critical thinking, technical research, documentation, and engineering evaluation in order to make informed decisions and implement the most appropriate solutions. For each category, I will apply relevant technical skills, including software security, UI/UX design, Java programming, and Object-Oriented Programming for software engineering and design; Data Structures, Algorithms, and complexity analysis for the data structures and algorithms category; and Cloud

Computing along with relational and document-oriented database technologies for the database category.

This deliverable aligns with the second outcome by requiring the development of professional-quality written documentation that clearly explains the research and implementation decisions. It also supports the first outcome by establishing clear development guidelines that encourage collaborative work practices within a computer science environment. Finally, it aligns with the third outcome by evaluating different technologies, analyzing their advantages and trade-offs, and applying computing principles to select the most appropriate solutions across the different technical categories.

iii. Skills and outcomes planned to be illustrated in the professional self-assessment

For the professional self-assessment, I will conduct a retrospective analysis to communicate who I am as a professional, what I have learned throughout my academic experience, and what I am now capable of accomplishing. This reflection will require critical thinking, self-awareness, and the ability to clearly express technical concepts in a professional manner. This deliverable aligns directly with the second outcome because it requires the development of professional-quality communication tailored to different audiences.